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Short Report

Diffusivities and atomic mobilities in fcc Pt–Al alloys

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ABSTRACT

Based on the first-principles computed activation energy, the frequency prefactor for impurity diffusivity of Pt in fcc Al was calculated by means of the Swalin correlation. The Vignes–Birchenall correlation originally developed for binary solutions with unlimited solubility was simplified to evaluate the interdiffusivities in binary terminal solutions with limited solubility. The simplified correlation was validated in fcc Ni–Al alloys, and then applied to evaluate interdiffusivities in fcc Pt–Al alloys, showing a reasonable agreement with the limited experimental data. On the basis of the limited experimental diffusivities in the literature and the presently evaluated interdiffusivities, a set of reasonable atomic mobilities in fcc Pt–Al alloys was established. This combinational approach by utilizing the limited experimental data, first-principles results, semi-empirical/empirical correlations and Diffusion Controlled TRANSformation (DICTRA) software package is of general validity and applicable in other binary alloys with limited diffusivity information.

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1. Introduction

The microstructure information is one of the most important factors determining the mechanical and physical properties of materials. A quantitative description of microstructural evolution in target alloys during various material processes is thus the core of high-performance alloy design. Among all the available computational techniques, the phase-field method is perhaps the most powerful and effective tool for microstructure simulation [1–3]. In addition to a quantitative phase-field model, the link to the real CALPHAD of PHase Diagram (CALPHAD) thermodynamic and atomic mobility databases is the prerequisite [4,5] for achieving the quantitative simulation of the microstructural evolution. So far, reliable thermodynamic databases for a wide variety of alloys have been constructed through about 30 years effort within the CALPHAD community [6,7]. However, this is not the case for the atomic mobility databases.

One major obstacle associated with the establishment of the atomic mobility database is that the experimental diffusivities are usually very limited. The situation is even worse for alloys with noble metals, e.g. Au, Pt, Pd, etc., which are too expensive for extensively experimental measurements. Nowadays, atomistic simulations [8–10], e.g. Molecular Dynamic (MD), Monte Carlo (MC) and First-Principles (FP), and some semi-empirical relations

[11] may sometimes serve as an important supplement to the experiments. However, the reliability of MD and MC simulations and the simple semi-empirical relations applied to real alloys always needs verification by experimental data. Though FP calculations can usually give reliable results, they are still limited to self- and impurity diffusivities. However, the reliable concentration- and temperature-dependent tracer and/or chemical diffusivities are indispensable to establishment of accurate atomic mobility database. As a consequence, there is an urgent need to develop an approach to acquire reliable diffusivities and atomic mobility databases in the systems with limited experimental information.

The Pt–Al system is one of such systems with extremely limited experimental diffusivities because of the noble element Pt. Furthermore, the Pt–Al system is an important sub-system in Ni-based superalloys and Thermal Barrier Coating (TBC) systems. Knowledge of its diffusivity is necessary for full understanding of microstructural evolution in Ni-based superalloys and TBC systems during service. Consequently, the major purpose of the present work is to establish a reasonable set of atomic mobility parameters in fcc Pt–Al alloys based on the limited experimental data available in the literature together with some theoretical methods and the Diffusion Controlled TRANSformation (DICTRA) software [12]. From such atomic mobilities coupled with thermodynamic information, various temperature- and composition-dependent diffusivities can be conveniently calculated.

The main body of this paper is organized as follows: all the diffusion information available in the literature for fcc Pt–Al alloys is critically reviewed in Section 2. After that, different kinds of

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theoretical methods for evaluating diffusivities and atomic mobilities are presented in Section 3. Moreover, a simplified Vignes–Birchenall correlation is also proposed to evaluate the interdiffusivities in binary terminal solutions with limited solubility in the section. In Section 4, the impurity diffusivities and interdiffusivities in fcc Pt–Al alloys are evaluated, and compared with the limited experimental data. Based on those evaluated interdiffusivities together with the limited experimental data, atomic mobilities in fcc Pt–Al alloys are then assessed by using the DICTRA software.

2. Literature review

For terminal fcc phase in the Pt–Al system, there exist only four pieces of information on impurity diffusivities and interdiffusivities in addition to certain amount of self-diffusivities in pure fcc Al and fcc Pt available in the literature. By means of the Pt/Pt–1.29 wt % Al diffusion couples combined with the electron microprobe analyzer (EMPA) technique, the impurity diffusion coefficients of Al in Pt were determined by Bergner and Schwarz [13] at 1373–1873 K. Later on, Lappalainen and Anttila [14] measured the impurity diffusion coefficients of Al in polycrystalline fcc Pt in the temperature range of 723–1173 K by employing Al-implanted Pt samples coupled with the Nuclear Resonance Broadening (NRB) technique. Based on the five-frequency rate model, activation energy for impurity diffusion of Pt in fcc Al was computed by Simonovic and Sluiter [15] using ab initio calculations coupled with Transition State Theory (TST) [16]. Unfortunately, the frequency prefactor was not determined by them due to the complex calculations. Instead, they assumed the frequency prefactor to be of the same order as that for Al self-diffusion [15]. Very recently, Eastman [17] thoroughly investigated the phase equilibria and interdiffusion in Ni–Cr–Al–Pt quaternary system using the diffusion-multiple approach in a combination with Scanning Electron Microscopy (SEM), Electron Backscattered Diffraction (EBSD), and EPMA techniques. The interdiffusion coefficients in fcc Pt-rich Pt–Al alloys were also measured [17] at 1283 and 1423 K via the Sauer–Freise method. The extrapolated impurity diffusivities at 1283 and 1423 K based on the corresponding interdiffusivities were found to agree reasonably with the data by Bergner and Schwarz [13]. Thus, both sets of impurity diffusivities by Bergner and Schwarz [13] and Eastman [17] are used to evaluate the atomic mobility for impurity diffusion of Al in Pt, while the data measured by Lappalainen and Anttila [14] at lower temperatures and in polycrystalline are not. Considering the fact that their results on similar elements, e.g. Ni and Au, agree well with the experimental values, the computed activation energy for impurity diffusion of Pt in fcc Al by Simonovic and Sluiter [15] can be regarded to be a reliable one. However, a more reasonable frequency prefactor than that for self-diffusion in pure Al is in need in order to evaluate the atomic mobility for impurity diffusion of Pt in Al. Besides, only the interdiffusion coefficients measured by Eastman [17] at two temperatures are not enough to evaluate an accurate set of atomic mobility parameters. More reasonable and self-consistent interdiffusivities are needed.

3. Methodology

3.1. Corrections between activation energy and frequency prefactor

Theoretical analysis indicates that there is a correlation between the activation energy Q and the frequency prefactor D_0 for impurity diffusion coefficients, which can be used to evaluate the frequency prefactor for impurity diffusion of Pt in Al based on

the first-principles calculated activation energy. According to Askil [11] and Du et al. [18], three such correlations are commonly used. The simplest correlation is due to Dushman and Langmuir [19],

$$D_0 = \frac{Qa^2}{Nh}, \quad (1)$$

where a is lattice constant, N Avogadro constant, while h Planck's constant. The second one is proposed by Zener [20] based on using elasticity theory and random walk theory,

$$D_0 = a^2 v \exp\left(\frac{\lambda\beta Q}{RT_m}\right), \quad (2)$$

where λ is the fraction of energy which goes into straining the lattice, and equals 0.6 for fcc and 0.8 for bcc metals. β is the dimensionless constant and equals $-d(\mu/\mu_0)/d(T/T_m)$ with μ and μ_0 elastic moduli at temperatures T and 0 K. The value of β is between 0.25 and 0.45 for most metals. R is gas constant while T_m is the melting temperature. The third semi-empirical correlation is derived by Swalin [21] according to electrostatic model using a Thomas–Fermi approximation,

$$\frac{\partial \log_{10} D_0}{\partial Q} = \frac{\alpha[(1+qr) + 0.75(-q^3r^3 + 6q^2r^2 + 5qr + 5)]}{2.3R[1 - 0.25(q^2r^2 - 5qr - 5)]}, \quad (3)$$

where α is the thermal expansion coefficients of the solvent, r the interatomic distance of the solvent, and q is the screening constant. Du et al. [18] employed both Zener and Swalin correlations to predict the impurity diffusion of some solutes in fcc Al, and found Swalin equation can describe more experimental data than Zener equation. According to Du et al. [18], the probable reason is that the electrostatic factor in Al is more important than the strain factor. Therefore, the Swalin correlation (Eq. (3)) is utilized to evaluate the frequency prefactor for impurity diffusion of Pt in Al based on the first-principles calculated activation energy [15].

3.2. Vignes–Birchenall relation

As stated above, most of the current first-principles calculations are limited to prediction of self- and impurity diffusivities. Though several tries were made to compute the interdiffusion coefficients using first-principles calculations [22,23], it is still far from the direct reproduction of the experimental data considering very complex diffusion mechanisms for interdiffusion in alloys. Fortunately, one empirical correlation between the concentration dependence of interdiffusion coefficient and deviation from the linearity of the solidus temperature was proposed by Vignes and Birchenall [24] for binary solutions with unlimited solubility,

$$\tilde{D}_{AA}^B = \phi(D_A^B)^{x_B}(D_B^A)^{x_A} \exp\left(-\frac{16\Delta T_s^{AB}}{RT}\right). \quad (4)$$

Here, D_A^B is the impurity diffusion coefficient of A in B, while D_B^A that of B in A. ΔT_s^{AB} represents the difference between the solidus temperature of the phase diagram and the temperature which would be computed if the solidus line varied linearly with composition. x_A and x_B are the concentrations of elements A and B. The factor 16 before ΔT_s^{AB} was found by Vignes and Birchenall [24] by trial and error, and is with the same dimension as gas constant R . ϕ is the thermodynamic factor, and defined as the following in binary system,

$$\phi = 1 + \frac{\partial \ln \gamma_A}{\partial \ln x_A} = 1 + \frac{\partial \ln \gamma_B}{\partial \ln x_B}, \quad (5)$$

where γ_A and γ_B are the activity coefficients of elements A and B. This Vignes–Birchenall correlation was also justified by some

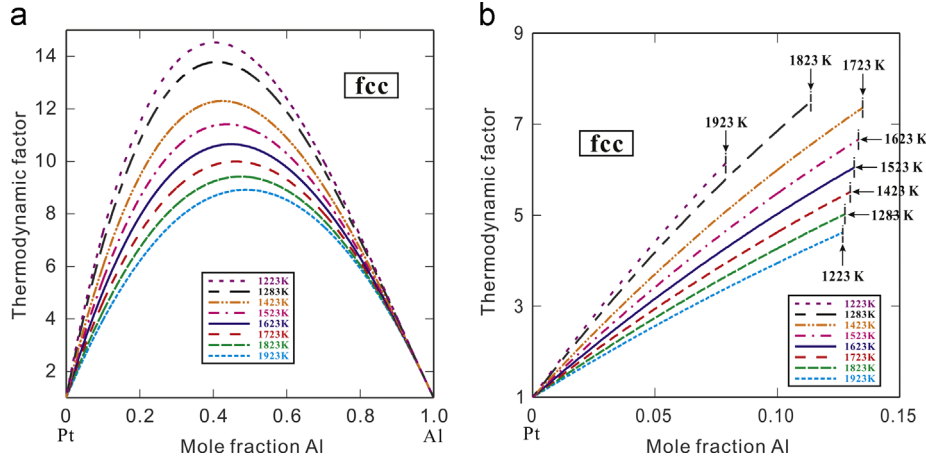


Fig. 1. (a) Calculated thermodynamic factor ϕ for fcc phase in the binary Pt–Al system over the entire composition range at different temperatures according to the thermodynamic description by Kim et al. [27], and (b) an enlarged part of thermodynamic factors at different temperatures over the corresponding stable composition ranges in Pt-rich region. The dashed vertical lines denote the stable phase boundaries for fcc phase in binary Pt–Al system at different temperatures.

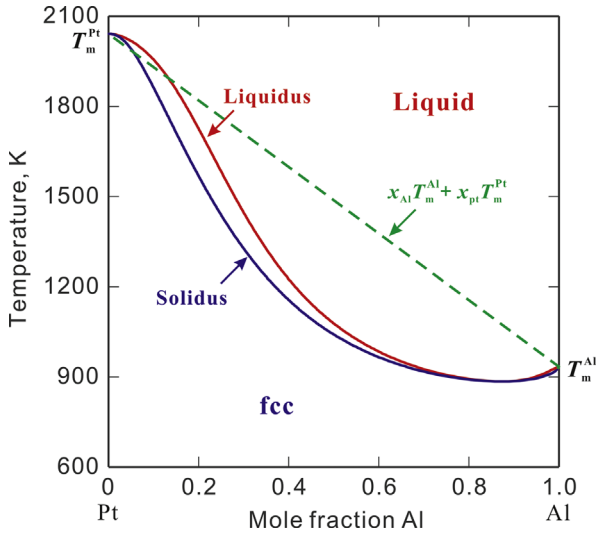


Fig. 2. Calculated metastable phase diagram including only liquid and fcc phases in the binary Pt–Al system. The dashed line representing $x_{\text{Al}} T_m^{\text{Al}} + x_{\text{Pt}} T_m^{\text{Pt}}$ is superimposed. Here, T_m^{Al} and T_m^{Pt} are melting points for pure Al and Pt.

empirical correlations between diffusivity and physical properties, and validated in a series of binary solutions with unlimited solubility by Vignes and Birchenall [24] and subsequent investigators [25,26].

3.3. Simplified Vignes–Birchenall relation for binary terminal solutions with limited solubility

However, the present fcc Pt–Al phase represents another class of binary solution, i.e. terminal solution with limited solubility. The stable terminal fcc solution phase can be extended into metastable region according to its Gibbs energy function, and then ϕ and ΔT_s^{AB} needed in Eq. (3) can be computed. Fig. 1(a) presents the calculated thermodynamic factor ϕ for fcc phase in the binary Pt–Al system over the entire composition range at different temperatures according to the thermodynamic description by Kim et al. [27]. An enlarged part of thermodynamic factors at different temperatures over the corresponding stable composition ranges in Pt-rich region is displayed in Fig. 1(b). Fig. 2 shows the calculated metastable phase diagram including only liquid and fcc phases in the binary Pt–Al system. The “solidus line” varied linearly with composition, i.e., $x_{\text{Al}} T_m^{\text{Al}} + x_{\text{Pt}} T_m^{\text{Pt}}$, is also superimposed in the plot. Here, T_m^{Al} and T_m^{Pt}

are melting points for pure Al and Pt. With the difference between the solidus line and the line of $x_{\text{Al}} T_m^{\text{Al}} + x_{\text{Pt}} T_m^{\text{Pt}}$, ΔT_s^{AB} can be thus obtained.

Before the direct usage of Eq. (3) in fcc Pt–Al alloys, one argument should be clarified. In such binary terminal solutions with limited solubility, the thermodynamic factor always equals 1 at the end sides of binary phase diagram (i.e. the pure elements). It changes abruptly when the concentration moves to the other side, and approaches to an extremum at a certain concentration. This feature of thermodynamic factor violates the general trend of interdiffusivities in a variety of terminal solutions with limited solubility range, e.g. fcc Ni–Al [28], fcc Cu–Al [29], etc. Instead, the general trend of interdiffusivities (logarithmic values) are rigorously related with ΔT_s^{AB} , which is the difference between the solidus and the temperature which would be computed if the solidus line varied linearly with composition. Therefore, one modification can be made to the original Vignes–Birchenall correlation, i.e. ϕ is roughly assumed to be 1 in the stable region of the terminal solution phase. Then, we propose a simplified Vignes–Birchenall correlation to evaluate the interdiffusion coefficients in binary terminal solutions with limited solubility,

$$\tilde{D}_{AA}^B = (D_A^B)^{x_B} (D_B^A)^{x_A} \exp\left(-\frac{16\Delta T_s^{AB}}{RT}\right). \quad (6)$$

If one uses the Arrhenius formula to replace the respective diffusivities, the following equations can be obtained,

$${}^0\tilde{D}_{AA}^B = ({}^0D_A^B)^{x_B} ({}^0D_B^A)^{x_A}, \quad (7)$$

$$Q_{AA}^B = x_B Q_A^B + x_A Q_B^A + 16\Delta T_s^{AB}. \quad (8)$$

Here, ${}^0D_A^B$, ${}^0D_B^A$ and ${}^0\tilde{D}_{AA}^B$ are the frequency prefactors for D_A^B , D_B^A and $\tilde{D}_{AA}^B$, while Q_A^B , Q_B^A and Q_{AA}^B are the respective activation energies. Based on Eq. (8), one knows that $16\Delta T_s^{AB}$ in fact acts the correction between the activation energy for interdiffusion and the mixture of activations energies for impurity diffusion.

The natural logarithmic value of interdiffusion coefficient $\ln \tilde{D}_{AA}^B$ can be further expressed as

$$\ln \tilde{D}_{AA}^B = x_B \ln D_A^B + x_A \ln D_B^A - \frac{16\Delta T_s^{AB}}{RT}. \quad (9)$$

Based on Eq. (9) (see also the formulae in the abstract of Vignes and Birchenall [24]), one can know that $\ln \tilde{D}_{AA}^B$ is a product of

a linear mixture of the natural logarithmic values for two impurity diffusion coefficients (i.e., $\ln D_A^B$ and $\ln D_B^A$) in a binary system and $-16\Delta T_s^{AB}/RT$, which conforms to the fact that the general trend of logarithmic values of interdiffusivities are rigorously related with ΔT_s^{AB} .

Before application in the present binary fcc Pt–Al alloys, Eq. (6) is validated in a similar system, e.g. fcc Ni–Al alloys, as presented in Fig. 3. That is because Ni and Pt belong to the same group. As shown in Fig. 3, the open circles represent the evaluated interdiffusion coefficients based on the presently simplified Vignes–Birchenall correlation, while the solid lines are due to our previous DICTRA calculations [28], which reproduce most experimental diffusivities well. It can be seen from the figure that the evaluated interdiffusion coefficients via the simplified Vignes–Birchenall correlation agree well with the DICTRA calculated ones. In addition, the very recently experimental interdiffusivities at 1283 and 1423 K in fcc Ni–Al alloys by Eastman [17] were also superimposed in Fig. 3. As can be seen, the experimental data due to Eastman [17] agree reasonably with both the evaluated interdiffusivities and the DICTRA calculated ones [28] if neglecting the scattering of the experimental data [17]. This fact indicates that the presently simplified Vignes–Birchenall correlation can give a reasonable prediction of the interdiffusion coefficients in such terminal solution phases with limited solubility, and thus can be applied to other similar binary systems, such as fcc Pt–Al alloys. Furthermore, the very recently experimental data from Eastman [17] should be also reasonable.

3.4. Atomic mobility

According to the absolute reaction rate theory [30] and the model suggested by Andersson and Ågren [12] and Jönsson [31], the atomic mobility for an element B in disordered fcc phase, M_B , can be expressed as

$$M_B = \exp\left(\frac{RT \ln M_B^0}{RT}\right) \exp\left(\frac{-Q_B}{RT}\right) \frac{1}{RT} \quad (10)$$

where M_B^0 is a frequency factor and Q_B is the activation enthalpy. In the spirit of the CALPHAD approach, the composition dependency of $RT \ln M_B^0$ and Q_B can be represented with the Redlich–Kister expansion, namely

$$\Phi_B = \sum_i x_i \Phi_B^i + \sum_{j>i} \sum_{l=0}^m x_i x_j \left[\sum_{r=0}^l r \Phi_B^{ij}(x_i - x_j)^r \right], \quad (11)$$

where Φ_B represents $RT \ln M_B^0$ or Q_B . Φ_B^{ij} is the value of Φ_B for pure i and $r \Phi_B^{ij}$ are binary interaction parameters. Obviously, one can simply combine Q_B and $RT \ln M_B^0$ into one parameter, i.e. $\Phi_B = -Q_B + RT \ln M_B^0$ in fcc alloys. Once the atomic mobilities in an alloy are known, various diffusivities, like trace and chemical diffusivities, can be calculated. Assuming the monovacancy mechanism for diffusion and neglecting correlation factors [12], the tracer diffusivity D_B^* is related to the mobility M_B by the Einstein relation,

$$D_B^* = RT M_B. \quad (12)$$

The only interdiffusion coefficient in a fictitious binary A–B system, $\tilde{D}_{AA}^B$, which relates the flux of element A with the gradient of component A and reference component B, is given by [12]

$$\tilde{D}_{AA}^B = x_A x_B \left\{ M_A \left(\frac{\partial \mu_A}{\partial x_A} - \frac{\partial \mu_A}{\partial x_B} \right) - M_B \left(\frac{\partial \mu_B}{\partial x_A} - \frac{\partial \mu_B}{\partial x_B} \right) \right\}, \quad (13)$$

where μ_A and μ_B are the chemical potentials of elements A and B.

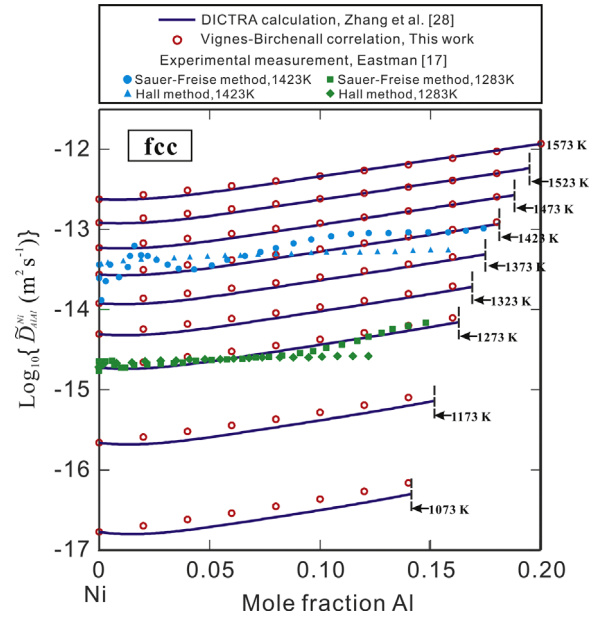


Fig. 3. Validation of the presently simplified Vignes–Birchenall correlation in fcc Ni–Al alloys by comparing the evaluated interdiffusivities due to the simplified Vignes–Birchenall correlation and those from the experiments [17] and the previous DICTRA calculation [28]. The dashed vertical lines denote the stable phase boundaries for fcc phase in binary Ni–Al system at different temperatures.

4. Results and discussion

The activation energy for impurity diffusion of Pt in fcc Al, (i.e. 144 kJ mole^{-1}) computed by Simonovic and Sluiter [15] was directly adopted in the present work, and the evaluated frequency prefactor for D_{Pt}^{Al} was evaluated as $2.210 \times 10^{-4} \text{ m}^2 \text{ s}^{-1}$ according to the Swalin correlation [21] (i.e. Eq. (3)). The impurity diffusivity of Al in fcc Pt, D_{Al}^{Pt} , was obtained by fitting the experimental data from Bergner and Schwarz [13] and Eastman [17] via the least-squares method. The obtained D_{Pt}^{Al} and D_{Al}^{Pt} were then employed to evaluate the interdiffusivities in fcc Pt–Al alloys by means of the presently simplified Vignes–Birchenall correlation, and the results are shown in Fig. 4. The phase equilibrium information is taken from the calculated phase diagram by Kim et al. [27]. In addition, the limited experimental interdiffusion coefficients measured by Eastman [17] are also superimposed in Fig. 4 for direct comparison. As can be seen, the evaluated interdiffusion coefficients due to the simplified Vignes–Birchenall correlation agree well with the experimental data [17] at 1423 K except for a slight deviation close to Pt side. One probable reason may be due to the fact that the high uncertainty usually occurs in diffusivity calculation using the Sauer–Freise method applied to the concentration limits approaching pure element [17]. That is because the slope of the composition profile at both sides is closed to zero, which causes the diffusivity to approach infinity. As for 1283 K, the evaluated interdiffusion coefficients due to the simplified Vignes–Birchenall correlation agree reasonably with the experimental data above $x_{Al}=0.03$. It should be noted that there exists a convex hull below $x_{Al}=0.03$, which shows large difference with the evaluated interdiffusion coefficients. There are two reasons for such convex hull. One reason lies in that the minimal random error can easily cause the concentration fluctuations, as pointed out by Eastman [17]. And the other is the above-mentioned feature of the Sauer–Freise method. Consequently, the evaluated interdiffusion coefficients according to the simplified Vignes–Birchenall correlation can be regarded to be reliable.

The above obtained diffusivities were then subject to DICTRA assessment to establish the atomic mobility database for fcc Pt–Al alloys. The thermodynamic parameters of fcc phase in Pt–Al

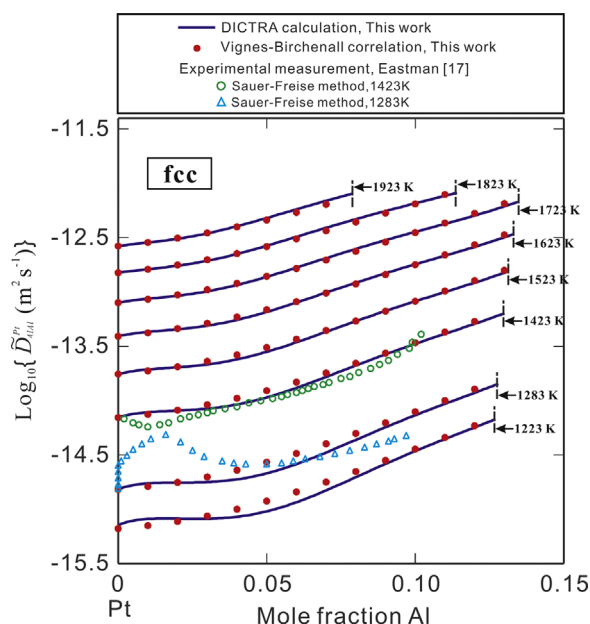


Fig. 4. Calculated interdiffusivities in fcc Pt–Al alloys based on the present atomic mobility parameters, compared with those due to the simplified Vignes–Birchenall correlation and the very recent experiments [17]. The dashed vertical lines denote the stable phase boundaries for fcc phase in binary Pt–Al system at different temperatures.

Table 1

Summary of atomic mobility parameters of Al and Pt in fcc Pt–Al alloys obtained in the present work.^a

Phases	Mobility	Parameters	Sources
fcc	Mobility of Al	$\phi_{Al}^{Al} = -123111.6 - 97.34T$	[28]
		$\phi_{Al}^{Pt} = -165348.4 - 154.81T$	This work
		${}^0\phi_{Al}^{Al,Pt} = +1021137.3$	This work
		${}^1\phi_{Al}^{Al,Pt} = +1582574.4$	
	Mobility of Pt	$\phi_{Pt}^{Al} = -144000.0 - 69.99T$	This work
		$\phi_{Pt}^{Pt} = -261426.9 - 99.17T$	[32]
		${}^0\phi_{Pt}^{Al,Pt} = -412386.1 - 455.04T$	This work
		${}^1\phi_{Pt}^{Al,Pt} = -1565135.2$	

^a Temperature (T) in Kelvin.

system due to the recent work by Kim et al. [27] were employed in the present work. The mobility parameters for pure fcc Al and fcc Pt were taken from Zhang et al. [28] and Liu et al. [32], respectively. The mobility parameter for impurity diffusivity of Al in Pt and that of Pt in Al were directly assigned as the presently obtained impurity diffusivities. The evaluated interdiffusion coefficients due to the simplified Vignes–Birchenall correlation, the experimental interdiffusivities at 1432 K, and those experimental ones above $x_{Al} = 0.03$ at 1283 K [14] are utilized to assess the interaction mobility parameters (Eq. (11)). The finally obtained atomic mobility parameters in fcc Pt–Al alloys are listed in Table 1.

The calculated interdiffusivities in fcc Pt–Al alloys according to the mobility parameters are also presented in Fig. 4, compared with the evaluated ones due to the simplified Vignes–Birchenall correlation and the experimental ones [17]. As can be clearly seen, the calculated interdiffusivities are in consistent with the evaluated ones, and also the reliable experimental data [17]. Fig. 5 shows the calculated self-diffusivity of Al and impurity diffusivity of Pt in Al, while Fig. 6 presents the calculated self-diffusivity of Pt and impurity diffusivity of Al in Pt along with the corresponding experimental data [13,14,17] according to the present atomic mobility parameters in Table 1. As can be seen in Fig. 6, the calculated impurity diffusivity of Al in Pt

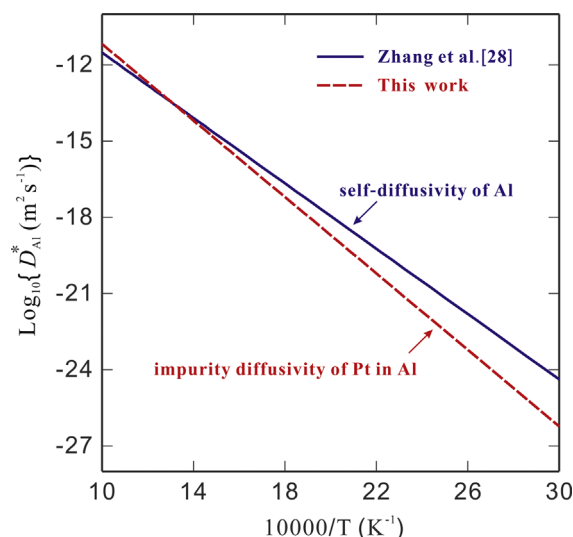


Fig. 5. Calculated self-diffusivity of Al [28] and impurity diffusivity of Pt in Al of temperature dependence.

agrees well with the experimental data by Bergner and Schwarz [13] and Eastman [17]. Moreover, the present calculations can also reproduce most of the experimental data from Lappalainen and Anttila [14] at lower temperatures though they were not used in the present assessment.

The presently established atomic mobilities in fcc Pt–Al binary alloys are to be included in the atomic mobility database for ternary Ni–Al–Pt alloys of technological importance by combining our previously published atomic mobilities in fcc Ni–Al [28] and Ni–Pt alloys [33]. In addition, the present combinational approach by fully considering the limited experimental data, first-principles results, semi-empirical/empirical correlations and DICTRA software package can serve as an effective method for evaluating diffusivities and atomic mobilities in other binary alloys with limited diffusivity information.

5. Summary

- The frequency prefactor for impurity diffusivity of Pt in fcc Al was evaluated on the basis of the Swalin correlation together with the first-principles computed activation energy.
- The Vignes–Birchenall correction was simplified in the present work in order to predict the interdiffusivities in binary terminal solutions with limited solubility. After validation in fcc Ni–Al alloys, the simplified correlation was then successfully applied to fcc Pt–Al alloys. The evaluated interdiffusivities in fcc Pt–Al alloys agree reasonably with the limited experimental data.
- Based on the limited experimental diffusivities available in the literature as well as the presently evaluated ones, a set of reliable atomic mobility parameters for fcc Pt–Al alloys was established via the DICTRA software.
- The present approach for successfully evaluating atomic mobilities in fcc Pt–Al alloys by using the experimental data, first-principles results, semi-empirical corrections and DICTRA in combination is of general validity and applicable to acquire diffusivities and atomic mobilities in other binary alloys with extremely limited experimental information.

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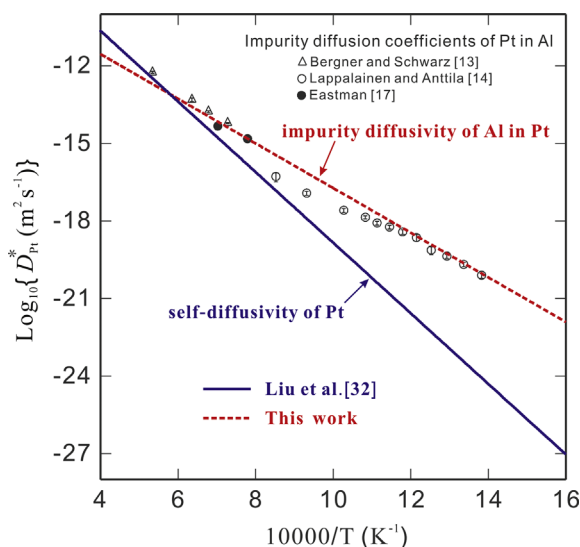


Fig. 6. Calculated self-diffusivity of Pt [32] and impurity diffusivity of Al in Pt along with the corresponding experimental data [13,14,17] according to the present atomic mobility parameters.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.calphad.2014.03.001>.

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